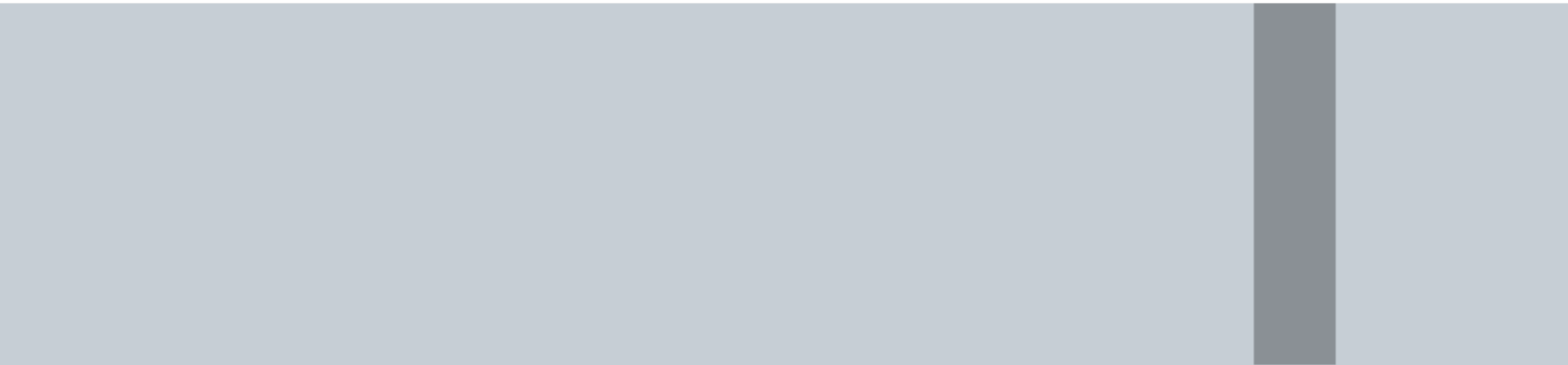
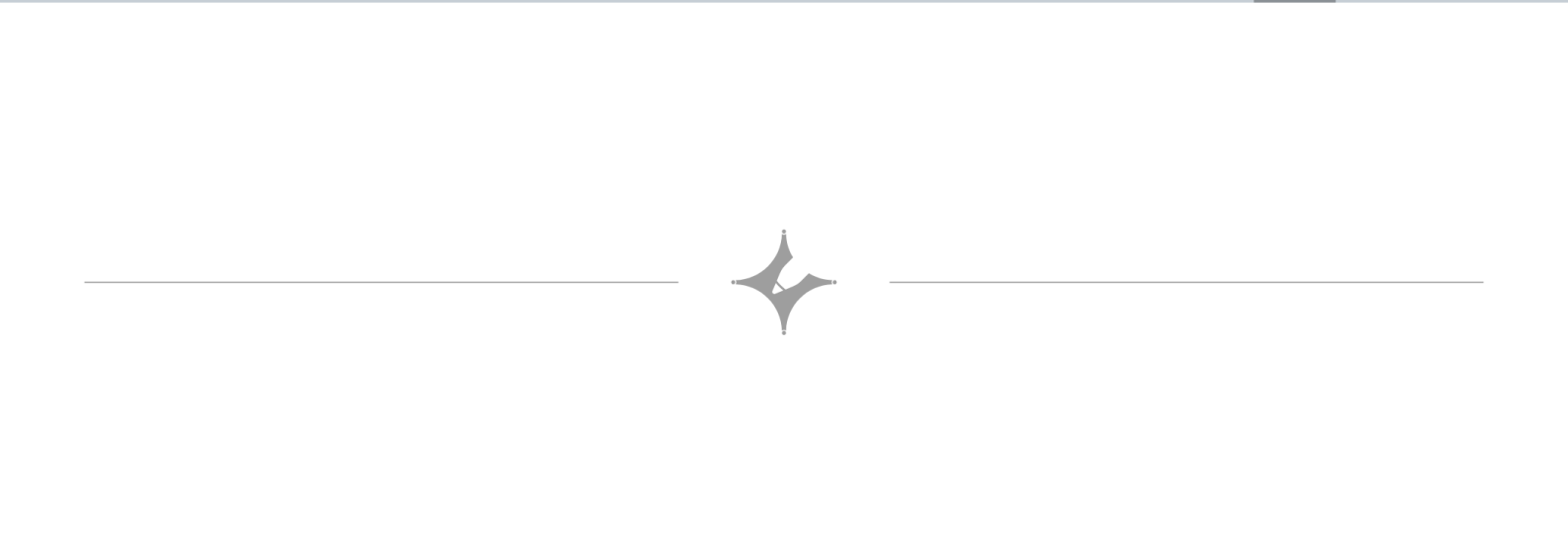
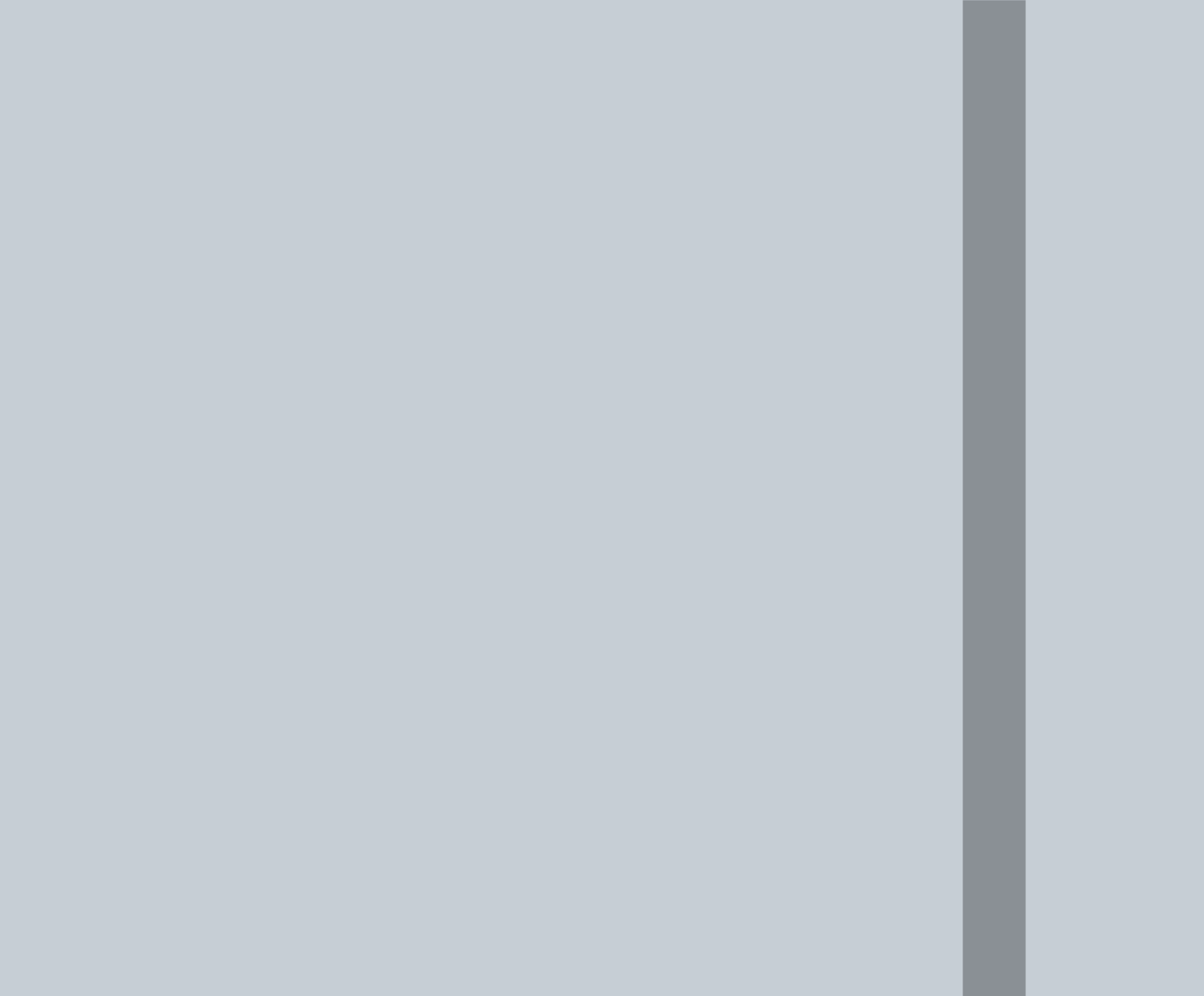




Profit and Loss

(1.) (G) Gain = Selling price - Cost price (SP - CP)
(G = SP - CP) (Some useful formulae)

(2.) L = CP - SP

(3.) $G\% = \frac{G}{CP} \times 100$, $L\% = \frac{L}{CP} \times 100$

Q6. Cost price of a toy was Rs 81. It is sold at a profit of 10% on selling price. Find selling price of the toy.

- A) Rs.72 B) Rs.99 C) Rs.100 D) Rs.90

SP = 100

G = 10% of 100 = 10

CP = 100 - 10 = 90

$\frac{CP}{90} = \frac{SP}{100}$

$1 = \frac{100}{90}$

$81 = \frac{100 \times 81}{90} = 90 \text{ ₹}$

OR

$81 + \frac{x}{10} = x$

$\Rightarrow \frac{10x - x}{10} = 81$

$\Rightarrow \frac{9x}{10} = 81 \Rightarrow x = \frac{810}{9} = 90 \text{ ₹}$

Q1. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?

- A) 30% B) 70% C) 100% D) 236%

Let CP = 100

G = 320% of 100 = 320

SP = 100 + 320 = 420

CP₂ = 125

G₂ = 420 - 125 = 295

G% w.r.t SP = $\frac{G}{SP} \times 100$

$= \frac{295}{420} \times 100$

$\approx \frac{300}{400} \times 100 = 75\%$

Ans = 70%

→ maine $\frac{295}{125} \times 100$ kara
→ CP

jo ki wrong hai (236% X)

SP ke respect me pucha hai to CP lenge niche..

Q3. Joey has 12 eggs with him. He sells x at a profit of 10% and remaining at a loss of 10%. He gains 5% on the whole. What is the value of x?

- A) 7 B) 9 C) 8 D) 10

12
x 10%
12-x 10%
+ 5%

$x \times \frac{110}{100} + (12-x) \times \frac{90}{100} = 12 \times \frac{105}{100}$

$110x + 12 \times 90 - 90x = 12 \times 105$

$20x = 12 \times 105 - 12 \times 90$

$20x = 12(105 - 90) = 12 \times 15$

$x = \frac{12 \times 15}{20} = 9$

Simple Interest (SI)

(P = Principle, Time = T (in yrs),
Rate = $R\%$ per annum)
 A = amount

$$SI = \frac{P \times R \times T}{100}$$

and

$$A = P + I \quad (I = \text{interest})$$

↓
Final amount

($\because I = A - P$)

Q1. In how many years will a sum double itself at 12.5% p.a. simple interest?

A) 4 ☒ B) 8 C) 10 D) 16

Let $P = 100$
 $A = 200$
 $SI = 100$
 $R = 12.5\%$
 $SI = \frac{PRT}{100}$

$A = P + I$
 $I = A - P$
 $T = \frac{SI \times 100}{P \times R}$
 $= \frac{100 \times 100}{100 \times 12.5} = 8$

Q3. A sum of Rs. 800 amounts to Rs. 920 in 3 years at SI. If the interest rate is increased by 3% it would amount to how much?

A) 992 B) 800 C) 900 D) 920

$P = 800$
 $A = 920$
 $T = 3y$
 $I = 120$
 $R = \frac{120 \times 100}{800 \times 3} = 5\%$

$SI_2 = \frac{800 \times 8 \times 3}{100} = 192$
 $A_2 = 800 + 192 = 992$

$Inc = 3\% \text{ of } 800 = 24$
 $Total\ Inc = 24 \times 3 = 72$
 $A_2 = 920 + 72 = 992$

Q6. A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is:

A. Rs. 650 B. Rs. 690 ☒ C. Rs. 698 D. Rs. 700

$A_3 = P + I_3 = 815 \quad \text{--- (1)}$
 $A_4 = P + I_4 = 854 \quad \text{--- (2)}$
 $(2) - (1)$
 $I_4 - I_3 = I_1 = 39$
 $I_3 = 39 \times 3 = 117$
 $\Rightarrow P = A_3 - I_3 = 815 - 117 = 698$

Compound Interest

Formula

If,
Principal = P
Rate = $R\%$ per annum
Time = T years

$$A = P \left(1 + \frac{R}{100}\right)^T$$

$$CI = A - P$$

$P = 100$
 $T = 3y$
 $R = 10\%$
 $CI = ?$

100
 $I \downarrow + 10\% = 10$
 110
 $II \downarrow + 10\% = 11 = 10 + 1$
 121
 $III \downarrow + 10\% = 12.1 = 10 + 1 + 1.1$
 133.1
 $CI = 133.1 - 100 = 33.1$

Q2. The compound interest on Rs.30,000 at 7% p.a. is Rs.4347.
The period (in years) is ____.

- A) 3 years B) 4 years ☒ C) 2 years ☒ D) 1 year

4347

$$\begin{array}{r} 30000 \\ \text{I } \downarrow +7\% = 2100 \\ \hline 32100 \\ \text{II } \downarrow +7\% = 2247 \\ \hline = 4347 \end{array}$$

Q4. A sum of money at compound interest doubled at a certain rate in 4 years. In how many years will it become 8 times at the same rate?

- A) 24 ☒ B) 12 C) 16 D) 18

$$P \xrightarrow{4y} 2P \xrightarrow{4y} 4P \xrightarrow{4y} 8P$$

$$4 + 4 + 4 = 12$$

Q6. What will Rs.2000 amount to in two years if it is invested in 20% p.a. compound interest, interest being compounded semiannually?

- A) Rs.2880 B) Rs.3160 ☒ C) Rs.2928.20 D) Rs.3148.40

$$\begin{array}{l} \text{I} \left\{ \begin{array}{l} 2000 \\ 6M \downarrow +10\% = 200 \\ \hline 2200 \\ 6M \downarrow +10\% = 220 \\ \hline 2420 \end{array} \right. \\ \text{II} \left\{ \begin{array}{l} 2420 \\ 6M \downarrow +10\% = 242 \\ \hline 2662 \\ 6M \downarrow +10\% = 266.2 \\ \hline 2928.2 \end{array} \right. \end{array}$$

Imp
It says semiannually
not annually